Bengali Image Caption Generation using Attention Mechanism

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***Abstract*—Image captioning has emerged as a rapidly thriving area for the machine learning research community. Generally, image captioning is performed by combining various computer vision features, natural language processing, and machine learning methods with consideration of some additional inputs to get more accurate context-dependent image captions *Bengali* is a significant language in India, adopted by approximately 100 million people. There exist various state-of-the-art methods for generating captions in the English language; however, for the Bengali language, there are very limited methods, and existing methods in the English language are not particularly helpful. Moreover, translations from English to Bengali may overlook or misinterpret subtle meanings, tones, or cultural nuances. Therefore, this work proposes a machine-learning model for captioning pictures in *Bengali* using an attention mechanism. The Flickr-8k dataset, which has 8000 images, is used to train the model in this work. The proposed method generates image captions in the *Bengali* language and attained a BLEU score of 0.66.**

***Index Terms*—Bengali, Image Captioning, Flickr-8k, LSTM, Beam Search, Seq2Seq Model**

I. INTRODUCTION

Image caption generation is a challenging task that produces a written description for an image. There are several uses for captioning images, such as captioning scenes for blind individuals [1], categorizing photos and videos according to different scenarios, improving the performance of image-based search engines, [2], answering questions visually, and context understanding [3]. Machine translation into Bengali can occasionally introduce errors, ambiguities, or unusual wording into English text. Even the most sophisticated translation algorithms face difficulties due to the complexity of grammar and the diversity of semantic difference among languages. These errors could detract from the overall quality and readability of the captions. When certain English terms or idioms have no exact equivalent in Bengali, the final Bengali caption may have awkward wording or inaccurate translations. Bengali and English may convey certain words and concepts extremely differently. By avoiding these semantic voids, direct generation guarantees that captions make sense in the context of the natural language flow. Most state-of-the-art methods for image captioning employ techniques that involve deep learning (DL) methods. One such approach consists of decoding an image using encoder-decoder methods, followed by utilizing a language model to convert the understanding of the image into words in a proper sequence. Natural Language Processing research has significantly benefited from using numerous DL methods in recent years, making it possible for computers to learn from, understand, categorize, and extract information from the world of words.

The most recent works on caption generation, use an encoder-decoder framework to produce the matching text for a single image. Several neural network topologies are employed as the encoder because of the source's diverse traits and behaviors. For instance, Recurrent Neural Networks (RNNs) are used for sequential data, whereas Convolutional Neural Networks (CNNs) are used for images. The attention technique considers the previously produced words in the target phrase to determine the relative relevance of the encoder's hidden states at each time step. In contrast, the attention mechanism lacks the ability to model globally and operates in a sequential fashion. A review network was developed to address this shortcoming, with review stages situated between the encoder and the decoder. As a result, more succinct, abstract, and thorough annotation vectors are produced, which later exhibit their utility in the sentence construction process.

This study presents an attention strategy for automatic image captioning using ResNet [4], VGG16 [5], and LSTM. One encoder-decoder framework was used in the design of the recommended framework. We employed a pre-trained CNN as the encoder, which converts a picture into a feature vector representing graphical features. LSTM was then selected as the decoder to provide the descriptive sentence. To increase performance, we included the Bahdanau [6] attention model, which enables learning to be concentrated on a particular area of the image.

The remainder of this paper is organized as follows: Section I discusses how image captioning works with different methods with attention mechanisms. Section II discusses various related studies. In Section III, the proposed methodology is elaborated. The results and the vast experiments are presented in Section IV and analyzed in Section V. Finally, the paper is concluded with some suggestions for future works in Section VI.

II. Related Work

Several works have conducted studies in this field and have suggested various techniques for characterizing the image using the Flickr-8k dataset. However, very few articles have explored multiple possibilities in image captioning using multi-modal datasets [7]. They have worked on image captioning in languages such as *Japanese*, *Hindi* [8], *German* [9], and *Arabic* [10]. *Nepali* [11] is built on top of the ML model and generates image descriptions in *Nepali*. Several works have used attention mechanisms in image captioning. Ami et al. [12] assessed their suggested model using a *Bengali* translation of the Flickr-8k dataset. Their goal was to confirm that visual attention could be used to create captions in *Bengali*. Using Google Cloud Translation, research created the Flickr-8k-Hindi [8] Datasets, a Hindi picture caption dataset based on Flickr-8k photographs. It has been demonstrated that paying attention to visuals is an effective way to generate captions for images [13], [14]. These attention-based captioning models may pick up on where in the image to focus when learning the target language. They could pick up on the spatial attention distribution in the CNN's [15] final convolutional layer or the semantic attention distribution from visual cues gleaned from social media photos. These approaches show how effective the attention [16] mechanism is, but they don't look into the contextual data that the encoding sequence contains. It entails producing succinct captions by the application of several methods, including Deep Learning (DL), computer vision [17], and natural language processing [18]. It unveiled a system that creates the captions using an encoder, a decoder, and an attention mechanism applied Attention on Attention (AoA) [19] to the image captioning model's encoder and decoder [20].

In this paper, we looked into the encoder-decoder framework to create image captions. Images have been used to represent works in both English and regional languages in the field of picture captioning.Here the image is encoded using a pre-trained CNN (ResNet and VGG16) by the encoder, and the captions are created iteratively by the decoder using LSTM. The proposed method for captioning images uses an attention mechanism that can identify fine-grained captions by focusing on the image's key components. Lastly, we employ the BanglaLekha and public flickr8k datasets to empirically verify the usefulness of the proposed method in producing image captions. We compare the BLEU score (0.66) of the proposed work with the BLEU scores produced by the model proposed in [6] and [21]. The attention layer in the proposed model is unique such that each hidden state of an encoding stage contributes to the construction of a decoding word. It is organized in a sequential order. We proposed an architecture for image captioning tasks with Xception as encoder and GRU [22] as decoder with an attention mechanism. To the best of our knowledge, the proposed work is the first where Xception is implemented as encoder and GRU as decoder for *Bengali* image captioning. Here the attention layer has an arrangement of sequence layers.

III. Methodology

This section thoroughly explains the proposed model along with a critical analysis of the implementation. After using several encoder-decoder models, we chose this model since it gave a better performance.

A. Dataset Description

A unique benchmark collection for sentence-based picture description and search is the Flickr-8k dataset. It has eight thousand photos and five distinct subtitles highlighting the main topics and events. The photographs were picked from six distinct Flickr groups and carefully chosen to show a range of settings and scenarios rather than any famous persons or places.

The file is structured as follows: pictures and captions, each separated by a new line that contains the image name, a space, and a CSV description of the image. The BanglaLekha Image Captions dataset, which consists of *Bengali* language image annotations, has also been utilized. The photographs have human annotations in *Bengali*. With *English* annotations, all widely used image captioning datasets exhibit a strong cultural bias toward the West. When using such datasets to train an image captioning system, it is assumed that the target culture was present in the original dataset and that there is a solid *English* to the target language translation system. $9,154$ photos make up the dataset that is being displayed. We hunted it down to $6,000$ images and used the dataset. We trained and tested our model on two datasets to validate its in-depth performance. It also helped us to analyze and compare the BLEU scores in a preferred manner.

B. Model Architecture

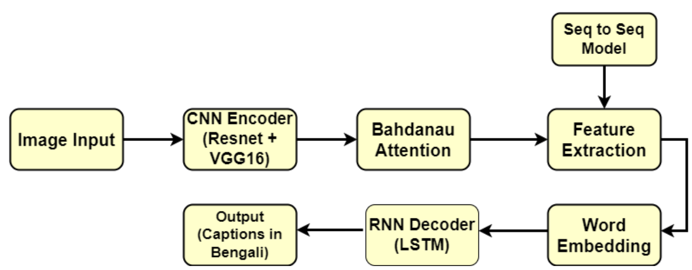

Fig. 1: Proposed model

We initialized the imports and loaded the util functions. After this, the helper functions to plot the tensor image are implemented. We defined the transformations to be applied, reshaping the images of the dataset to 228. As depicted in Fig.1, our proposed model used the seq2seq model [23]. In the encoder, pre-trained ResNet model [4] is used to extract the features. Decoder is the implementation of the Bahdanau Attention Decoder [6]. We use both the Fickr-8k and BanglaLekha datasets as input. We employed LSTM for the attention decoder model. The LSTM [24]is utilized with 252 channels, and the dropout layer is set to 0.25. The remaining parameters are: embedding dim = 100, buffer size = 128; batch size = 64; and optimizer used is Adam [25]. Following that, we use 50 epochs to train the proposed model.

1) CNN Encoder: We make use of convolutional neural networks [15], a subset of deep neural networks that are often used for assessing visual images. It employs a special technique called convolution. As it is currently understood, convolution is a mathematical operation on two functions that results in a third function that characterizes the transformation of one's shape by another. We employ it to simplify the images without sacrificing any of the essential characteristics needed to provide a reliable prediction. We design ResNet [26] architecture by implementing various ResNet architectures using the Tensorflow [27] and Keras API. Our approach makes use of the BanglaLekha and Flickr-8k datasets. We adjust various hyperparameters needed for the ResNet design. In order to prepare our dataset for training, we also performed some preprocessing on it. The learning rate is adjusted based on the total number of epochs. To guarantee improved learning, the learning rate must be lowered as the number of epochs increases. We also employed VGG16 [5] since it demonstrated a notable improvement over the state-of-the-art configurations by evaluating the networks and increasing the depth using an architecture with very small convolution filters.

2) Bahdanau Attention: For Attention Modelling, we used Bahdanau Attention [6]. This method differs from the standard encoder-decoder primarily in that it does not try to encode a whole input sentence into a single fixed-length vector. Rather, it converts the input text into a series of vectors and, in the process of decoding the translation, adaptively selects a subset of these vectors. The suggested approach looks for a group of locations in a source sentence where the most pertinent information is concentrated before producing a term in a translation. Based on all of the previously created target words as well as the context vectors connected to these source positions, the model then predicts a target word. To optimize the Attention decoder, we employ four hidden layers [16].

3) Feature Extraction: To deal with such a large amount of data, feature extraction [28] is needed. It is a step in the dimensionality reduction process, which divides and condenses a starting collection of raw data into smaller, easier-to-manage groupings. These huge data sets' abundance of variables is by far their most significant feature. Processing these variables takes a lot of computer power. By choosing and merging variables into features, feature extraction efficiently reduces the amount of data by assisting in the extraction of the best feature from these large data sets. We employ a seq2seq model [23] for this. It creates an output word sequence by taking an input word sequence. For this, we use a seq2seq model. It takes an input of a sequence of words and generates an output sequence of words. It use a recurrent neural network to achieve this. Since RNN has a vanishing gradient issue, we turned to LSTM [24] instead. By using two inputs at any given time one from the user and one from its previous output it builds the word's context. We can see the implementation of the seq2seq paradigm in Fig. 2. The encoder translates the input words into matching hidden vectors by utilizing deep neural network layers. Every graphic depicts both the word in use and its context. The encoder and the decoder

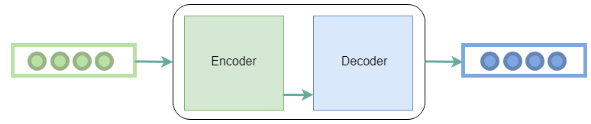

Fig. 2: Seq2Seq model used in this work

are comparable. To create the next hidden vector [29] and ultimately forecast the next word, it uses the encoder's hidden vector, its own hidden states, and the current word as input. The decoder can selectively examine the input sequence thanks to the application of the attention technique. The decoder identifies the word with the greatest probability as the output. Variable-length sequences in a seq2seq framework are enabled by the use of zero padding to both the input and output. Thus, we use the concept of bucketing.

4) Word Embedding: We employed GloVe [30] as a word embedding model to produce a map of the word values in order to give vector value embedding. It functions as a neural network algorithm that assigns a probability value and chooses the value with the higher probability. The fundamental premise of GloVe word embedding is to ascertain the association between words using statistical research. The co-occurrence matrix, unlike the occurrence matrix, denotes the frequency of simultaneous appearances of a certain word pair. Each value in the co-occurrence matrix represents a pair of words that co-occur. In contrast, the continuous n-skip gram algorithm uses the current word as its input and makes accurate predictions about the words that will come before and after it. Every word in this study had 100 dimensions, and each word's vector space measured 306×100.

5) RNN Decoder: In order to caption images, we developed an LSTM [24] based model that uses feature vectors from the VGG network to predict word sequences, or captions. In recent picture caption generation tasks, the models based on recurrent neural networks and deep convolutional networks have dominated. Performance and intricacy are timeless subjects.

By combining the benefits of LSTM with simple RNN, which outperforms a dominated one in terms of performance and efficiency. The suggested method separates the RNN's hidden units into many identically sized components and allows them to operate concurrently. Their outputs are then combined with matching ratios to produce the final outputs. Better results than the dominant structure are obtained by conventional training on the Flickr-8k dataset, without the need for additional training data. The reason for using LSTM as the decoder is that it solves the issue of exploding and vanishing gradients that come from the backward propagation step.

C. Setting of Hyperparameter

To avoid the problem of high variance, there was a need to prevent overfitting. These models have limited generalizability. This is a significant challenge for image captioning. When we looked at our model's performance, we saw that overfitting, not underfitting, was the problem. Our model has undergone

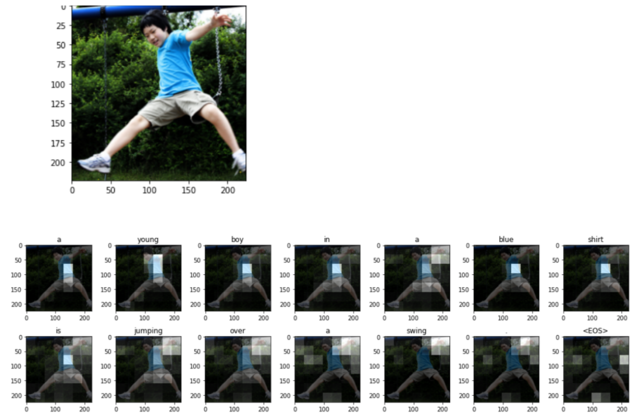

Fig. 3: Different Attention regions generated by the proposed method

some hyperparameter [31] tweaking in order to reduce the overfitting issue.

To do this, various dropout settings have been applied to the decoder and sequence model image features. It involved an initial phase which started with baseline parameters and baseline feature engineering [32]. Then we focused on warm up phase where we played around manual tuning or grid search of a few important parameters. Lastly, we went for intense tuning phase on more parameters and final feature engineering [32] dropouts help prevent overfitting. For several fully connected layers, different activation functions were used.

IV. Results

The GloVe [30] word embedding is used for text preprocessing, the Bahdanau Attention Mechanism [6]is used for weighting vector features, and the LSTM is used for decoding in our model. Our model was trained over 50 epochs, and for each epoch, we obtained the train loss and validation loss. Fig. 3 shows Bahdanau Attention [6] implemented on a shuffled image selected from the Flickr-8k dataset. We can say that the model is perfectly trained and the Validation loss increases while the epochs are increasing. Fig. 4 shows the pooling layers of the encoder used during feature extraction from the images. We used BLEU score criteria to assess the proposed model. Based on the images and evaluation scores, the majority of the captions utilizing our model had the highest accuracy and syntactic correctness.

TABLE I: BLEU scores obtained using Flickr-8k dataset

Search	**Learning Rate**	**Model Used**	**BLEU-1**	**BLEU-2**	**BLEU-3**
Beam=0	0.0001	Resnet	0.465	0.505	0.447
		VGG16	0.357	0.495	0.418
		Resnet	0.576	0.485	0.237
		VGG16	0.612	0.499	0.298
Beam=1	0.0001	Resnet	0.243	0.465	0.533
		VGG16	0.287	0.368	0.478
		Resnet	0.541	0.472	0.185
		VGG16	0.414	0.247	0.178
Beam=2	0.0001	Resnet	0.143	0.225	0.288
		VGG16	0.665	0.568	0.548
		Resnet	0.241	0.172	0.155
		VGG16	0.424	0.117	0.128

```
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
input_1 (InputLayer)         [(None, 224, 224, 3)]     0
_________________________________________________________________
block1_conv1 (Conv2D)        (None, 224, 224, 64)      1792
_________________________________________________________________
block1_conv2 (Conv2D)        (None, 224, 224, 64)      36928
_________________________________________________________________
block1_pool (MaxPooling2D)   (None, 112, 112, 64)      0
_________________________________________________________________
block2_conv1 (Conv2D)        (None, 112, 112, 128)     73856
_________________________________________________________________
block2_conv2 (Conv2D)        (None, 112, 112, 128)     147584
_________________________________________________________________
block2_pool (MaxPooling2D)   (None, 56, 56, 128)       0
_________________________________________________________________
block3_conv1 (Conv2D)        (None, 56, 56, 256)       295168
_________________________________________________________________
block3_conv2 (Conv2D)        (None, 56, 56, 256)       590080
_________________________________________________________________
block3_conv3 (Conv2D)        (None, 56, 56, 256)       590080
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Fig. 4: Extraction of features from images

Different methods are incorporated in this proposed mode, like ResNet, VGG16, and LSTM. As a result, Table I shows the single embedding architecture's performance on the Flickr-8k dataset. It is evident that the suggested model outperformed the alternative models [6], [21]. After training the job from the above configurations, the captions were catalyzed. Fig. 5 shows the generated captions in *Bengali* and *English* from the BanglaLekha Image Captions dataset. We plotted the loss vs epochs graph for the BanglaLekha Image Captions Dataset, as shown in Fig. 6.

V. Performance Analysis

The proposed model is compared on BLEU scores with existing models Inception3 + GRU [6] (Model-I) and CNN + LSTM [21] (Model-II) as shown in Table II. It is evident that the performance provided by the proposed model is superior. In contrast to the model [6] that was suggested, which utilized Inception3 and GRU [22] and [21] whose model used CNN and LSTM, we employed Resnet, VGG16, and LSTM in our model, which produced higher BLEU scores.

Fig. 5: Generated captions from images

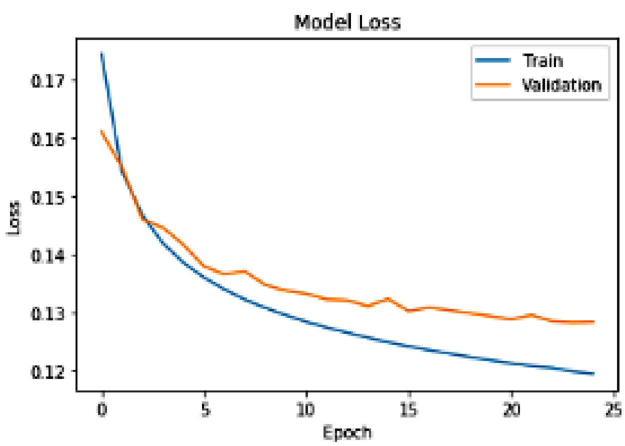

Fig. 6: Plotted graph of loss vs epochs

We initialized VGG16 and Resnet in our model using the Flickr-8k and BanglaLekha dataset weights, and we utilized a very tiny learning rate to fine-tune it from the start. This model outperforms one with a larger size kernel, which is why we chose it when many stacked smaller size kernels are used with a given receptive field. This is so that it may learn more complicated characteristics at a lesser cost. Several non-linear layers improve the network's depth. 3X3 kernels aid in maintaining the image's finer-level characteristics.

VI. Conclusion and Future Scope

Bengali image captioning entails not only data availability, a crucial factor, but also other obstacles including linguistic complexity and semantic issues. In this work, an image

TABLE II: Comparison of proposed model with existing models

Dataset	Model	BLEU-1	BLEU-2	BLEU-3
BanglaLekha Image Captions	Model-I [6]	0.286	0.355	0.294
Bengali Visual Genome (BVG)	Model-II [21]	0.412	0.499	0.398
Flickr8k + BanglaLekha	Proposed model	0.662	0.602	0.576

captioning system in *Bengali* is presented. The Bahdanau Attention is used in the suggested way to produce *Bengali* image captions. The Flickr-8k and BanglaLekha Image Captions datasets are used to train the model. Compared to other models, the proposed model received higher BLEU values. In order to increase the BLEU score, additional research on this model can be done in the future by training it with more explanations for each image. It is necessary to create a comprehensive and diversified dataset of *Bengali* captions in order to raise the scores. Nonetheless, the experimental findings indicate that the model can further the field of *Bengali* picture captioning study, and the proposed model may prove valuable in subsequent studies.

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